Twenty People and a Posting Board — Practice AS · 9618 §2.1 · Paper 1

Name: _____ Due: _____

What the schemes pay for: a **LAN** covers a *small geographical area* on *privately owned* infrastructure at a *high data transfer rate* — never "a local area"; a **difference** names *both* sides · a **drawing** of a star has *every device the stem lists, every one labelled, each connected only to the central device* · a packet on a star goes *to the switch*, which *checks the destination address* and sends it *only to the receiver* · a **client-server** answer names the server *and* the client *in the scenario* and says what each device does · a **thin or thick** client is described by *where the data is and where the processing is*, and a converse pair scores once · a **peer-to-peer** drawback needs its *reason*. Questions 1 and 2 are real Paper 1s, verbatim; 3 and 4 are ours and say so. Question 5 is the switch: **do part (a) with a pen before you open the file**.

1. June 2023 Paper 12 Q1(a)(b)(c)(e), verbatim. A company has a LAN (local area network).

(a) Give **two** benefits of connecting computers to a LAN. [2]

1 _____

2 _____

(b) Give **two** characteristics of a LAN. [2]

1 _____

2 _____

(c) One of the company's offices has one server and four computers connected in a star topology. Draw a diagram to show the layout of the office's star topology. [2]

(e) The network runs as a thick-client model. Describe what is meant by a thick-client model. [2]

2. November 2023 Paper 12 Q7(b), verbatim. A Local Area Network (LAN) contains four devices: a router, two laptop computers and a server.

(i) The LAN is set up as a star topology. Draw a diagram of the topology of the LAN. [2]

(ii) Explain how data is transmitted between the two laptops in the LAN. [2]

3. Ours — and it says so. A school opens a second campus on the other side of the city. The computers on both campuses are connected as a WAN (wide area network). On the main campus, the five servers in the server room are connected to each other in a mesh topology.

(a) State **two** ways in which a WAN is different from a LAN. [2]

(b) Describe what is meant by a mesh topology. [2]

(c) Explain **one** reason why a mesh topology, rather than a star topology, is suitable for the server room. [2]

4. Ours — and it says so. A food-delivery company has an app. A customer uses the app on a smartphone to see the menu of a restaurant and to place an order. The company's system is a client-server model.

(a) Describe the roles of the different devices in this client-server model. [4]

(b) Three students share their revision notes by setting up a peer-to-peer network between their laptops. Describe **one** drawback to the students of using a peer-to-peer network. [2]

💻 **5. The Switch — `challenge_U14a_TheSwitch.py`.** [2 + 3 + 1 = 6] How does the switch know which cable the receiver is on? Nobody tells it — it learns from the frames that pass through it. You write that switch.

(a) With a pen, before you touch the computer. [2] Six frames arrive at a five-port switch, in this order: **`(1, A→B)`** · **`(2, B→A)`** · **`(3, C→A)`** · **`(1, A→C)`** · **`(4, D→E)`** · **`(5, E→D)`** — the number is the port the frame arrived on. The rule: *learn* the sender's address on the port it arrived on; a *known* destination goes out of that one port only; an *unknown* destination goes out of every port except the one it came in on. Complete the table.

frame	arrived on port	the table after this frame (MAC → port)	sent out of port(s)
1 · A → B	1		
2 · B → A	2		
3 · C → A	3		
4 · A → C	1		
5 · D → E	4		
6 · E → D	5		

(b) [3] Write **`Switch.receive`** until the judge prints **6 / 6**. The fifth check runs the six frames above and expects the flooding to stop as the table fills; the sixth unplugs A and moves it to another port.

(c) [1] Run **`python3 challenge_U14a_TheSwitch.py bus`**. Write down how many of the six frames collided on the bus, and why the star had none.

🕳️ **Dig, if you dare:** in the challenge, add a seventh frame `(5, "F", "E")` — a new device F that shares port 5 with E (a small hub hangs off that port). Run `trace` with the extra frame, or step it with your own switch: which ports does the frame go out of, what does the table hold afterwards, and where does the next frame `(1, "A", "F")` go? Explain why two addresses can live on one port.

🚀 **Above and beyond — not in the total.** In the deck's `arp -a` capture the last line was `(192.168.220.255) at ff:ff:ff:ff:ff:ff` — the *broadcast* address, a frame addressed to *everyone*. Work out what your learning switch does with a frame whose destination is `ff:ff:ff:ff:ff:ff` (and why it must never learn that address as a *source*). Then the twist: the table in the hook was built by ARP, which asks "who has 192.168.220.4?" by broadcasting — so even the star, the shape that never broadcasts, needs a posting board for one job. Which job, and why can it not be done any other way?

Additional space. If you need more room for any answer, continue here and write the question number in the margin beside it.